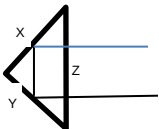


Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7 HT	(a)	(i)		Ultraviolet / UV (1) Microwaves (1)	2			2		
		(ii)		They all travel in a vacuum / at the same speed [in a vacuum] / all undergo reflection / refraction / travel in straight lines. Accept diffraction / interference / polarisation / transverse waves Don't accept transfer energy	1			1		
	(b)	(i)		 The vertical ray XY (by eye) (1) YZ roughly parallel with the incident ray and passing <u>straight</u> outside the prism (1)		2		2		
		(ii)		Angle [of incidence] needs to be greater than the <u>critical angle</u> (or 36°) / it is smaller than the <u>critical angle</u> (1) - for it to be <u>totally internally reflected</u> / it otherwise gets <u>refracted</u> [out of the back of] the prism] (1) Accept converse argument	2			2		
		(iii)		$c = f\lambda = (5.639 \times 10^{14}) \times (532 \times 10^{-9}) = 2.9999 \times 10^8 \text{ [m/s]}$ (1) $d = s \times t$ (1 – manip) ecf for c $= 2.9999 \times 2.5626 = 7.6876 \times 10^8 \text{ [m]}$ (1) Distance to moon = $3.8438 \times 10^8 \text{ [m]}$ (1) NB1. If 3×10^8 is used without calculation shown then first mark is lost. NB2. If workings are shown and 2.9999 is rounded up to 3 then all marks are available. (Answer is then 3.8439×10^8)		4		4	4	
		(iv)		So there is no overlap between the outward and return pulses Accept they don't interfere with or interrupt each other		1		1		
				Question 7 total	5	7	0	12	4	0